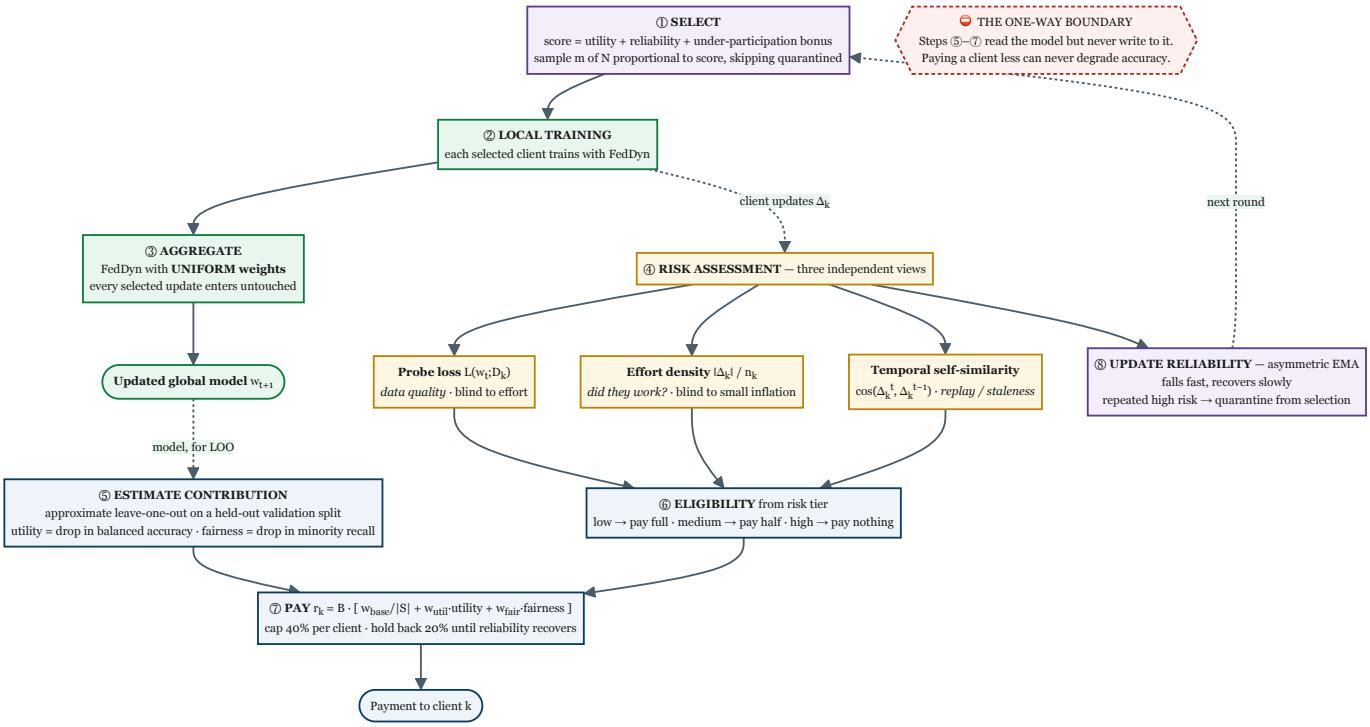
DCFI-UPA — proposed methodology and dataset coverage

Companion to `DCFI_TECHNICAL_REPORT.md`. The diagrams below render natively on GitHub.

1. The proposed system: one communication round

The design principle is a **strict one-way boundary**. The reliability signal may decide *who gets paid*; it may not decide *whose update enters the model*. That separation is what makes the incentive layer free of accuracy cost — it was measured, not assumed (see §4 and the technical report §9.3).



Why the boundary is one-way

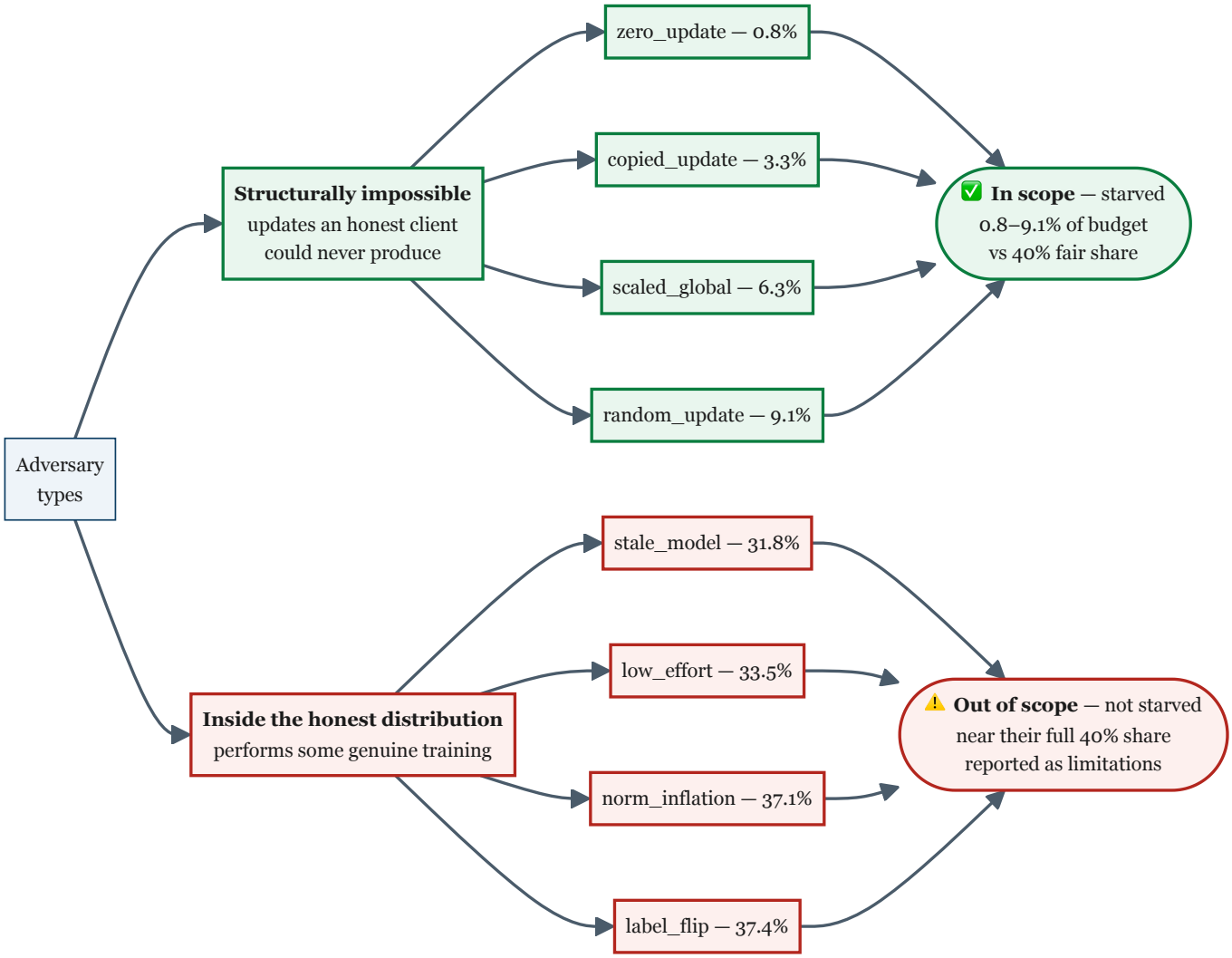
Our earlier method let the contribution signal set **aggregation weights**. A controlled ablation showed the better that signal worked, the *worse* the model became: the variant that detected nothing matched FedDyn (0.695 vs 0.697), and the variant that detected well lost 0.030. A label-noise client still holds usable gradient signal, and FedDyn's dynamic regularisation already bounds its damage — so down-weighting discards information for protection already present.

Measured effect of moving the gate (5 seeds, balanced accuracy):

Scenario	FedDyn	Gate on aggregation	Gate on reward only
clean	0.768	0.735	**0.737**
zero_update	0.768	0.647	**0.752**
copied_update	0.777	0.643	**0.762**
random_update	0.765	0.637	**0.750**

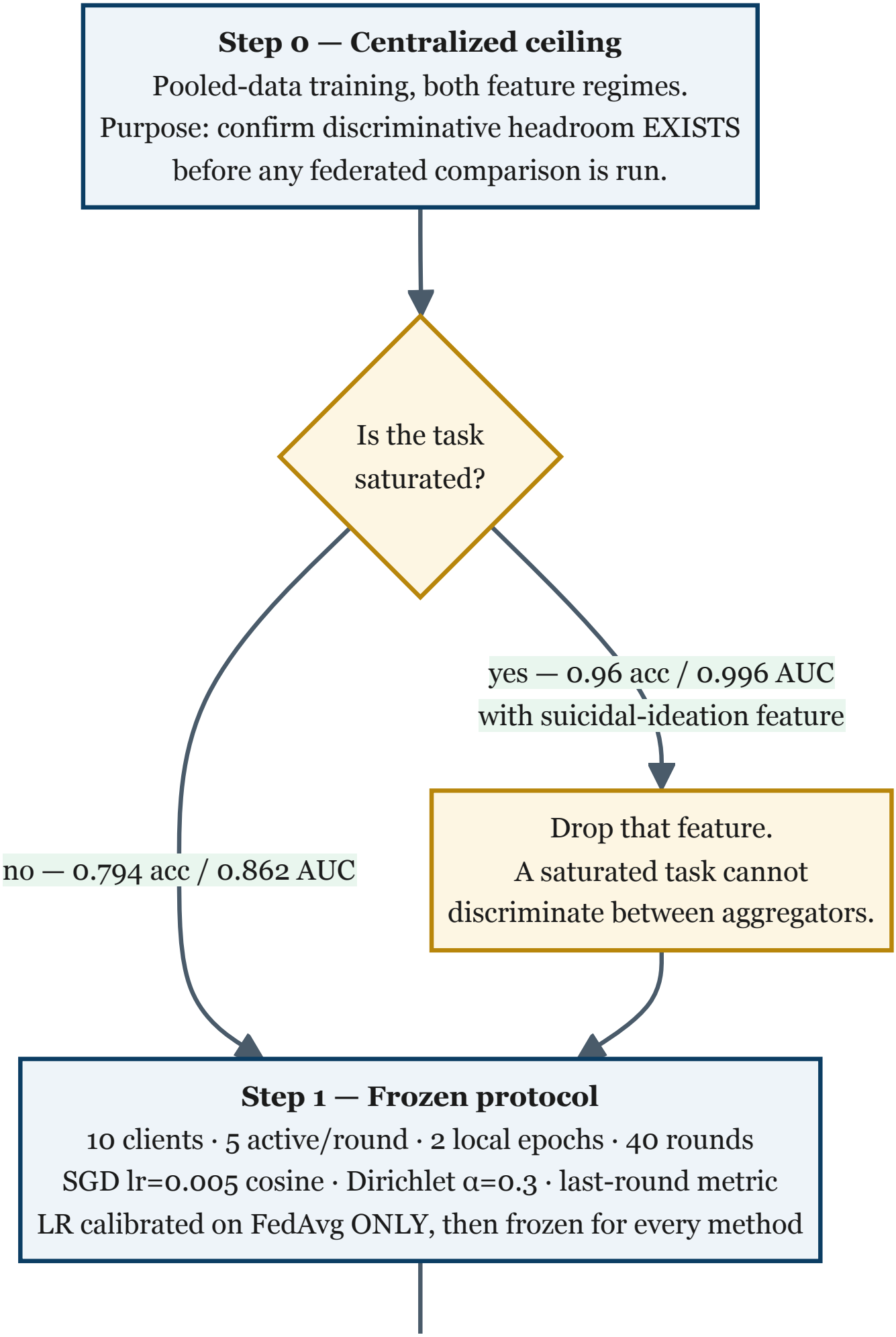
Adversary reward capture under reward-only gating stays at 0.00% / 1.97% / 6.80% against a 40% fair share.

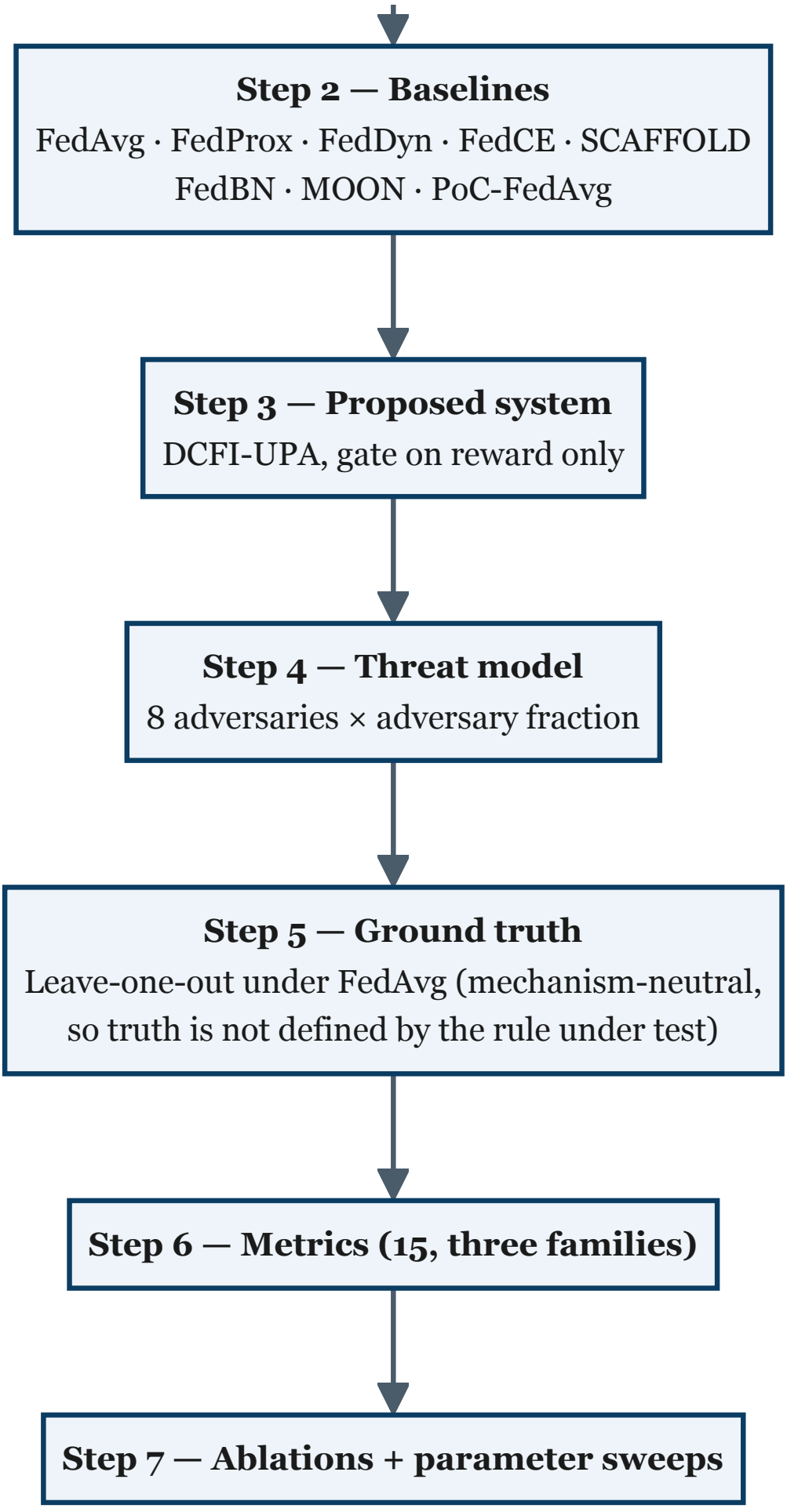
2. Threat model



The split is not arbitrary and it is not a convenience. An adversary that does *some* real training produces an update lying inside the natural spread of honest behaviour, and §7.6 of the technical report quantifies why that is unfixable by magnitude scoring: honest effort density has log-sd 1.014, so a 3× inflation sits 1.08 sd inside normal variation. Separating at 2 sd would demand inflation beyond 7.6×, which no attacker is obliged to use.

3. Experimental methodology


Step 0 — Centralized ceiling
Pooled-data training, both feature regimes.
Purpose: confirm discriminative headroom EXISTS
before any federated comparison is run.
Is the task saturated?
yes — 0.96 acc / 0.996 AUC
with suicidal-ideation feature
Drop that feature.
A saturated task cannot
discriminate between aggregators.
no — 0.794 acc / 0.862 AUC
Step 1 — Frozen protocol
10 clients · 5 active/round · 2 local epochs · 40 rounds
SGD lr=0.005 cosine · Dirichlet α=0.3 · last-round metric
LR calibrated on FedAvg ONLY, then frozen for every method



The fifteen metrics

Performance	Reward fairness	Fairness
balanced accuracy	Spearman ρ(reward, LOO)	Jain — useful-client rewards
macro-F1	reward–contribution MAE	worst-client balanced accuracy
AUC	reward share, positive vs negative contributors	selection-frequency Jain
convergence rounds	reward-to-cost ratio	ρ(reward, client size)
collapsed seeds	individual-rationality rate	minority-class recall

4. Datasets tried

Used in reported results

Dataset	Modality	Task	Runs	Where it appears
Student Depression (Kaggle `hopesb`, 27,901 rows)	tabular survey	binary depression screening, 58.6% positive	**869**	**Primary — all current DCFI-UPA results**
OCTMNIST	retinal OCT	4-class	38	Earlier robustness study; the CSSV negative result
ISIC / DermaMNIST	dermoscopy	7-class	20	Cross-dataset generalisation
APTOS-2019	fundus photography	5-class ordinal (QWK)	20	Cross-dataset generalisation
Depression Student (Kaggle `ikynahidwin`, 502 rows)	tabular survey	binary	ceiling only	Saturation check — showed 0.96 acc / 0.996 AUC with the suicidal-ideation feature, which is why the screening regime exists

Coverage: 4 datasets across 2 modalities with full federated results, plus one used as a saturation control. The core negative result (contribution-weighted aggregation does not beat FedDyn) replicates on all four.

Present in the codebase, not used for results

`chestxray14` (2 exploratory runs only), `cifar10`, `mnist`, `fmnist`, `brats` — loaders exist and work, but none are part of any reported study.

Why the tabular mental-health dataset is primary

- Matches the incentive setting: small institutional cohorts, self-reported and therefore noisy labels, genuine privacy motivation.
- Cheap enough to run 864 federations — the leave-one-out ground truth alone needs 390, which is infeasible on ResNet-scale imaging.
- Non-saturated in the screening regime, so aggregators are actually distinguishable (verified in Step 0 before any comparison was run).

Honest limitation

All current DCFI-UPA results are on **one dataset**. The earlier negative result is 4-dataset; the incentive mechanism is not yet. Porting DCFI to OCTMNIST or APTOS is the clearest way to strengthen external validity, and the code path is dataset-agnostic — it needs runs, not new implementation.